Lecture 3.1: AWS Compute

Maharishi International University

Department of Computer Science

M.S. Thao Huy Vu

Maharishi International University - Fairfield, Iowa

All rights reserved. No part of this slide presentation may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying or recording, or by any information storage and retrieval system, without permission in writing from Maharishi International University (MIU).

Agenda

- Overview of AWS Compute Services
- Amazon EC2 (Elastic Compute Cloud)
- Security Group
- Amazon Lightsail

Overview of AWS Compute Services



Overview of AWS Compute Services

- Virtual Machine (EC2)
- Lightweight and Simplified servers (Lightsail)
- Serverless Computing (Lambda)
- Containers (ECS and EKS)

Overview of AWS Compute Services

- Cloud Compute: Computing resources: Servers, Storage, APIs
- Benefits
  - Auto-scaling
  - Pricing options
  - Integration with other services
  - Security: isolation, compliance certifications, built-in encryption
  - Performance metrics: CPU/GPU
  - Network enhancements

Overview of AWS Compute Services

- Choose the right Compute option:
  - Computational power
  - Budget
  - Development complexity
  - Operational control
  - Practical application of services

Amazon EC2 (Elastic Compute Cloud)

- EC2 is a VPS (Virtual Private Server) on the cloud
- Users can use EC2 to build a network of servers
- Flexibility
- Scalability

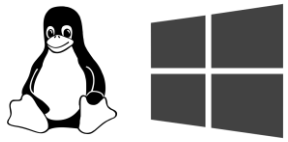
Computer and EC2 Instance



Computer



EC2 Instance



Operating System
AMIs
(Linux or Windows)



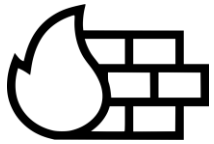
CPU & RAM
Instance Type



Hard Drive
EBS



Network Adapter
ENI



Firewall
Security Groups

Amazon EC2 (Elastic Compute Cloud)

- Pricing model
  - **On-Demand** Instances: Pay by the second, with no long-term commitments or upfront payments
  - **Reserved** Instances: Up to 75% discount compared to On-Demand, based on a one-year or three-year commitment
  - **Spot** Instances: Purchase unused EC2 capacity at significant discounts but with the possibility of termination by AWS if it needs the capacity back.
  - **Savings** Plans: Flexible pricing model that provides savings of up to 72% on specified usage in exchange for a commitment to a consistent amount of usage (measured in $/hour) for a 1 or 3-year period.

Amazon EC2 (Elastic Compute Cloud)

- Key features
  - Instance Types
  - Amazon Machine Images (AMIs)
  - Elastic IP Addresses
  - Security Group: Inbound, Outbound rules

IP Addressing

An IP address is the EC2 instance address on the network.

Private IP Address:

- EC2 instance receives the private IP from the subnet.
- All EC2 instances (all devices in the network) have a private IP address.
- Private IP addresses allow instances to communicate with resources in the same network.
- No cost

Public IP address:

- All EC2 Instances can be launched with or without a public IP address.
- Public IP addresses are required for the instance to communicate with the Internet.
- It is dynamic as it is changed when restarting the instance
- No cost

Elastic IP

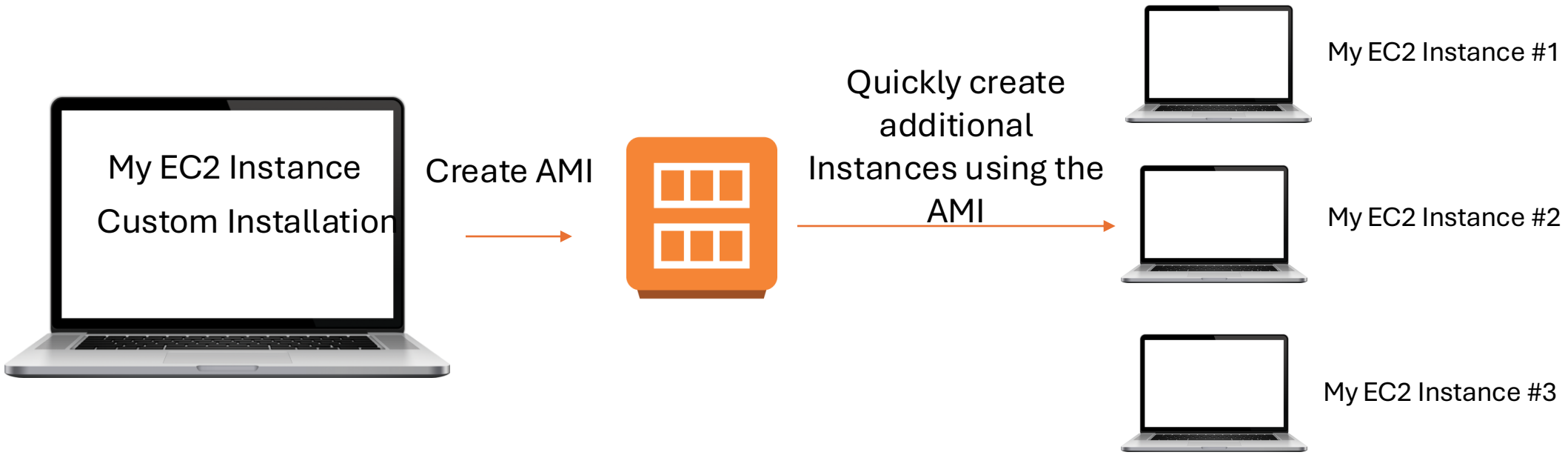
- A static, fixed and public IPv4 IP.
- Able to attach it to one instance at a time.
- Have up to 5 elastic IPs per region.
- Cost per one Elastic IP: $0.005 per hour, approx. $4 a month.

Amazon Machine Image - AMI

- Predefined information that are needed to launch an EC2 instance
- Includes:
  - OS
  - Software packages
  - Other required settings: Application code…

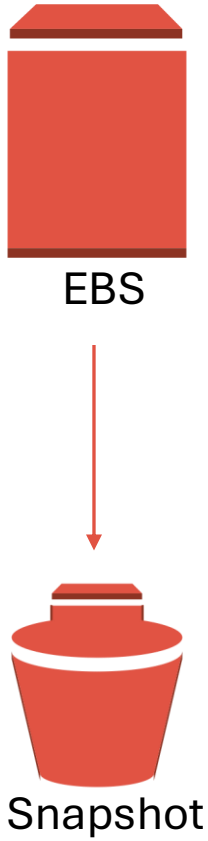


Understanding AMI



Snapshots

- An **image** of an EBS volume that can be stored as an **incremental backup** of the volume or used to create a duplicate.
- Not an active EBS volume. You cannot attach or detach a snapshot to an EC2 instance.
- To restore a snapshot, you need to create a new EBS volume using the snapshot as its template.



AMI vs. SAnapshots

AMI	**EBS Snapshot**
An entire EC2 instance definition that includes all EBS snapshots plus some metadata like kernel, AMI name, description, block device mappings, and more.	Backup of a single volume.
Could be used to deploy your applications on different machines easily.	You need to recover and mount a snapshot to an EC2 instance.

Instance Types

- The Instance Type determines the underlying hardware such as CPU, GPU, RAM, network and disk read/write capacity.
- Each instance type offers different compute, memory, and storage, graphic capabilities and is grouped into instance families based on these capabilities.
- NVIDIA instances have a GPU (accelerated computing) that works best for ML since a GPU provides high parallel and simultaneous computing.
- The same service price varies depending on the region. For example, the same instance in the USA costs higher than the one in Japan.

Read more: Instance Types

General-purpose and memory-optimized instance type

- **General-purpose** instances provide a balance of compute, memory, and networking resources. It can be used for web servers and code repositories. General-purpose instance types start with **T** and **M**.
- **Memory-optimized** instances are good for applications that process large amounts of data in memory such as caching. Memory-optimized instance types start with **R** and **X**.

Compute-optimized instance type

- Compute-optimized instances are ideal for compute-bound applications that benefit from high-performance processors. It is good for batch processing workloads, media transcoding, high-performance web servers, high-performance computing (**HPC**), scientific modeling, dedicated gaming servers, ad server engines, machine learning inference, and other compute-intensive applications. Compute-optimized instance types start with **C**.

Accelerated computing instance type

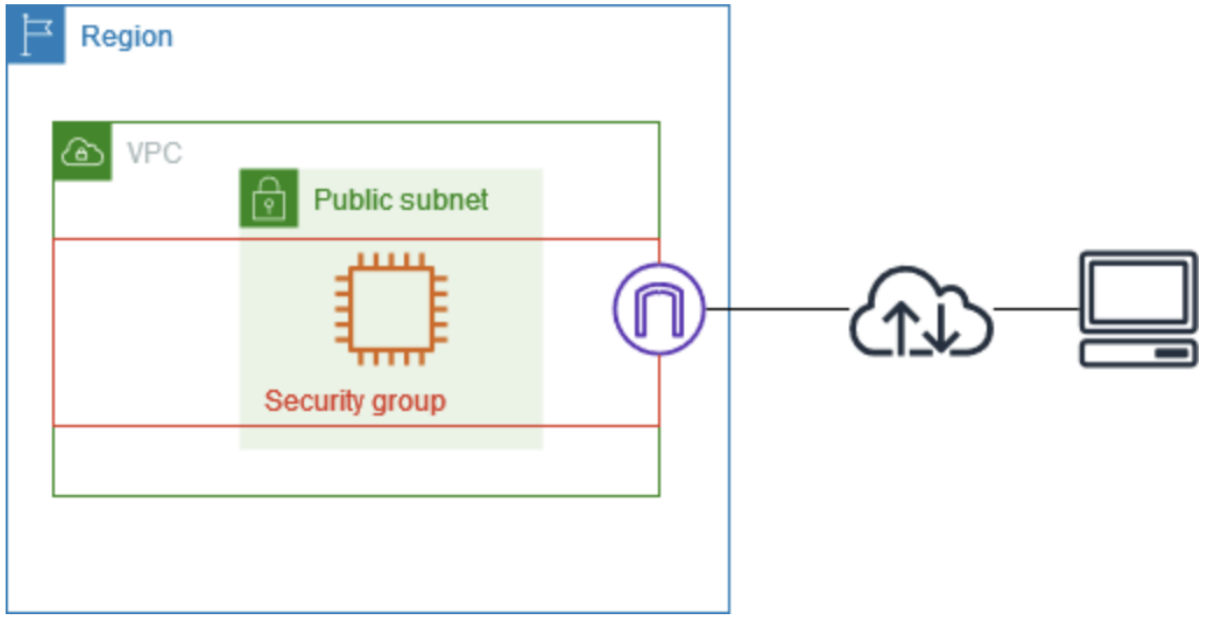
- Accelerated computing instances use hardware accelerators (GPUs) or co-processors to perform functions such as floating point number calculations, graphics processing, or data pattern matching.
- Both accelerated computing and compute-optimized instance types provide high computing power. But the way how they provide the computing power is different. Accelerated computing instances have GPUs which are accelerators whereas the compute-optimized ones have good high-performance processors. Accelerated computing instance types start with **P** and **G**.

Storage-optimized instance type

- Storage-optimized instances are designed for workloads that require high, sequential read and write access to very large data sets on local storage. They are optimized to deliver tens of thousands of low-latency, random I/O operations per second (IOPS) to applications. Storage-optimized instance types start with **I** and **D**.
- Do not configure EC2 as a database! Use Database Service which provides instance types designated for databases.

Security Groups

- A virtual firewall for EC2 instances to control inbound and outbound traffic
- Inbound Rules: Specify the type of incoming traffic to reach the instance
- Outbound Rules: Specify the type of outgoing traffic to leave the instance



Inbound/Outbound Rules

Name	Group ID	Group Name	VPC ID
	sg-b63da5fa	default	vpc-af9b48d5

Description | **Inbound** | Outbound | Tags

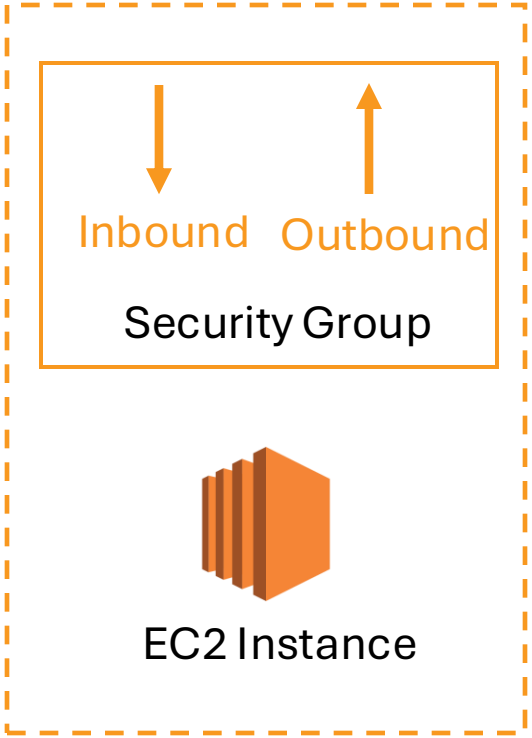
Edit

Type	Protocol	Port Range	Source	Description
All traffic	All	All	your IP	

Description | Inbound | **Outbound** | Tags

Edit

Type	Protocol	Port Range	Destination	Description
All traffic	All	All	0.0.0.0/0	

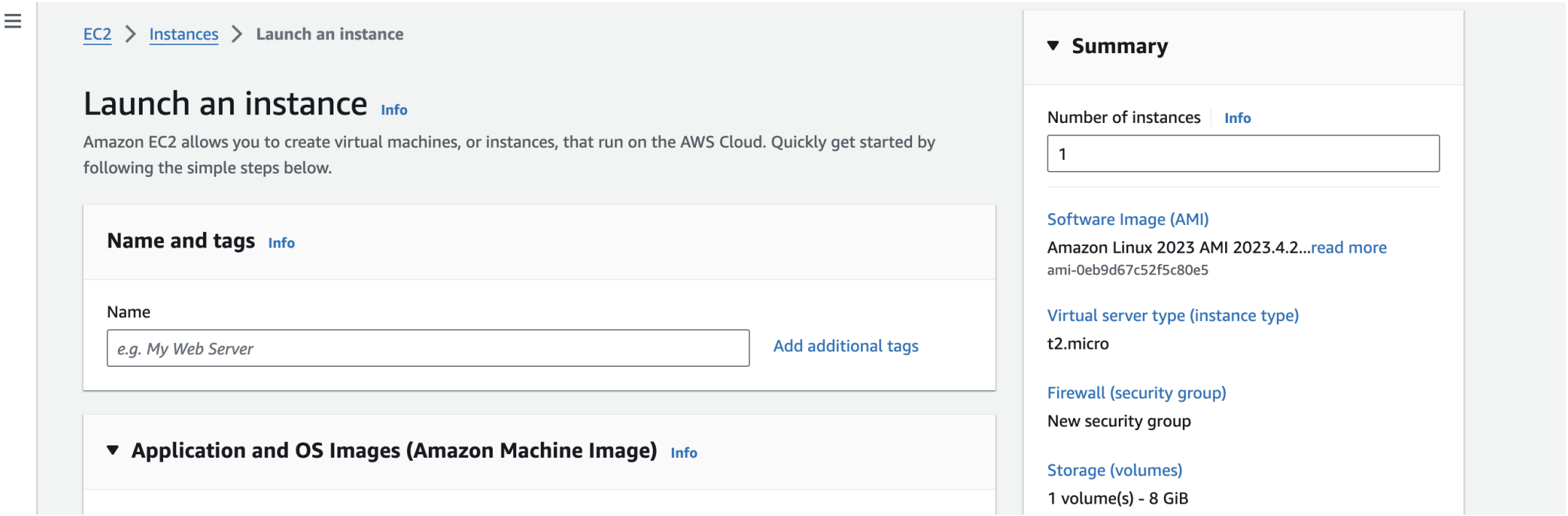


Virtual Private Network (VPC)



Amazon EC2 (Elastic Compute Cloud)

- Demo

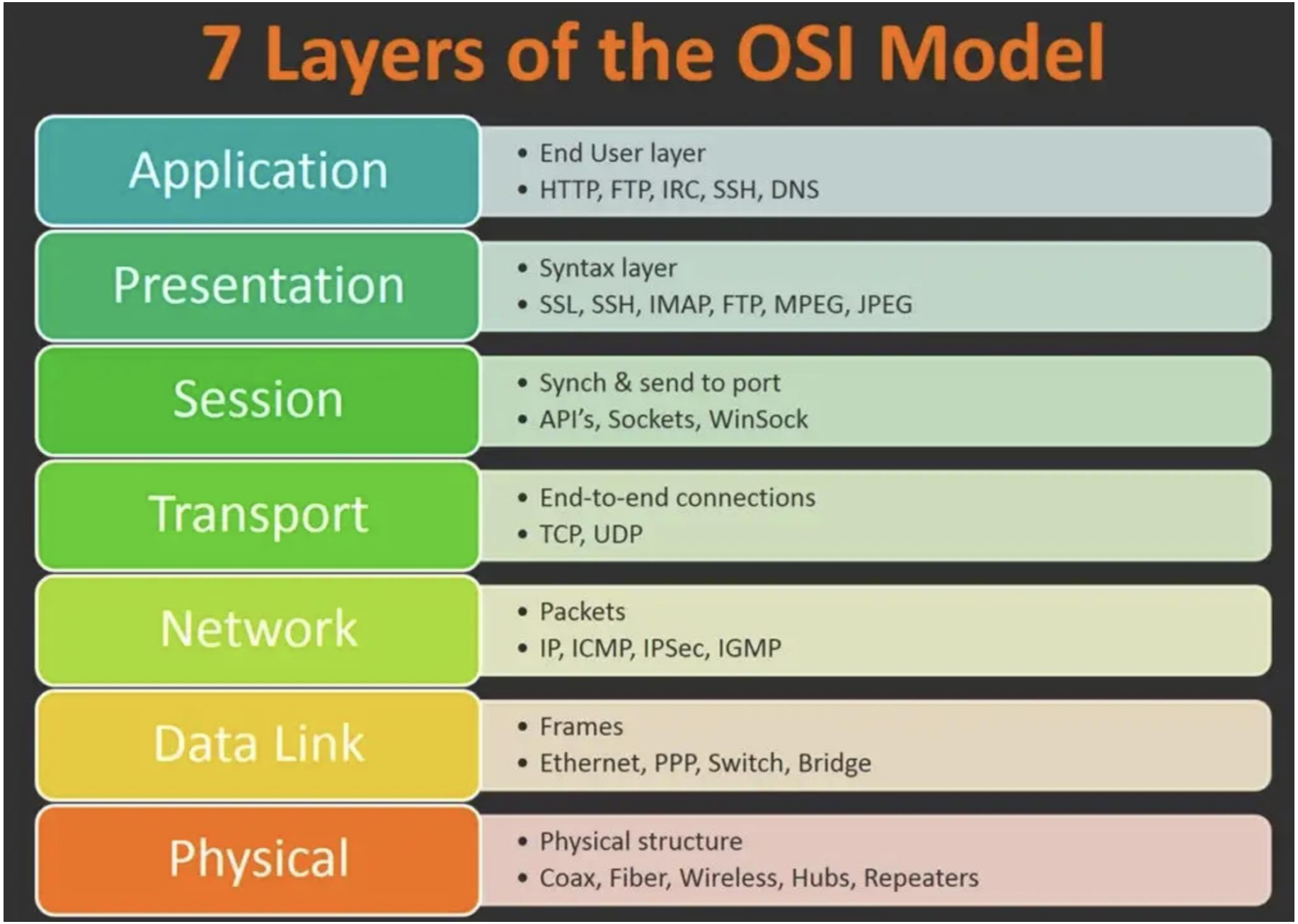


Elastic Load Balancer (ELB)

- A service that **distributes** incoming web **traffic** (visitors to a web site) and equally across multiple EC2 instances that are running the same app.
- It enables **fault tolerance** by seamlessly providing the required amount of load-balancing capacity needed to route application traffic. It helps prevent one server from being overloaded while another server can handle more visitors. It makes the app more reliable.
- ELB scaling and high availability is managed by AWS. You just create the load balancer and it can handle millions of transactions in a second. It will never be a bottleneck. But you have to manage the scalability of your app e.g., in EC2.
- Elastic Load Balancer is a general term. Underneath, there are multiple types of load balancers in AWS.

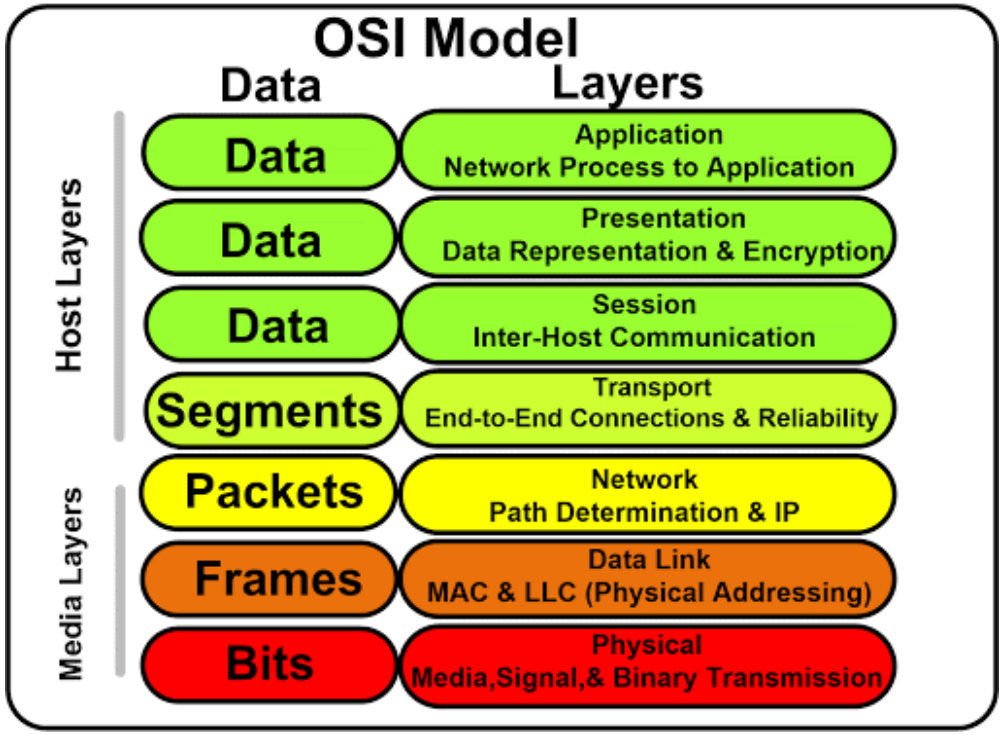
ELB

- Types of Load Balancer: Application Load Balancer, Network Load Balancer, or Classic Load Balancer
- Load Balancer configuration: Health checks, SSL/TLS
- Integration with Auto Scaling Group


7 Layers of the OSI Model
Application
• End User layer
• HTTP, FTP, IRC, SSH, DNS
Presentation
• Syntax layer
• SSL, SSH, IMAP, FTP, MPEG, JPEG
Session
• Synch & send to port
• API's, Sockets, WinSock
Transport
• End-to-end connections
• TCP, UDP
Network
• Packets
• IP, ICMP, IPSec, IGMP
Data Link
• Frames
• Ethernet, PPP, Switch, Bridge
Physical
• Physical structure
• Coax, Fiber, Wireless, Hubs, Repeaters

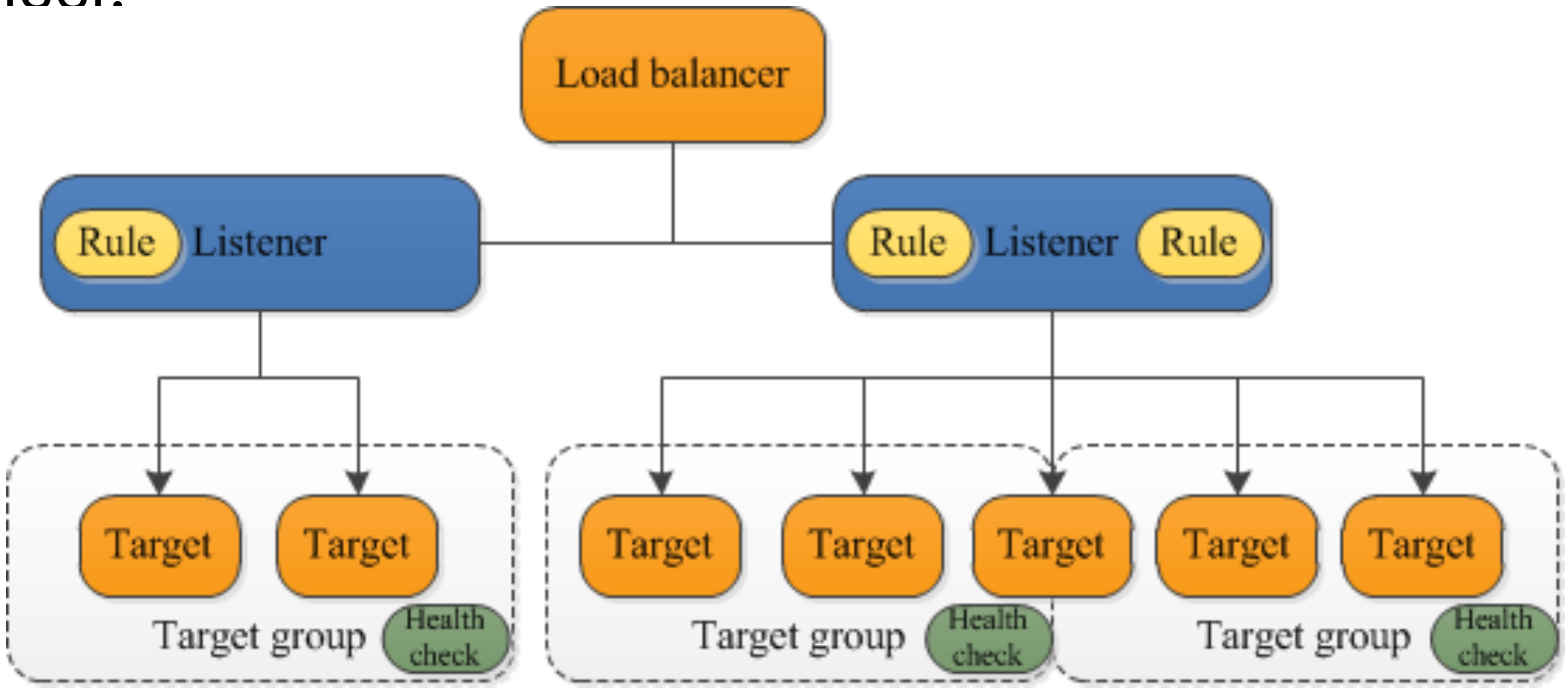
Types of ELB

1. **Application Load Balancer** (ALB) – Routes based on the content of the request (layer 7). Supports path-based, host-based, query string, parameter-based, and source IP-based routing. Supports IP addresses, Lambda, and containers as targets.
2. **Network Load Balancer** (NLB) – Routes connections based on IP protocol data (layer 4). Provides ultra-high performance and low latency. Supports UDP and static IP addresses as targets.
3. **Gateway Load Balancer** – Operates at Layer 3 (Network). Gateway Load Balancers enable you to deploy, scale, and manage virtual appliances, such as firewalls, intrusion detection and prevention systems, and deep packet inspection systems.
4. **Classic Load Balancer** (CLB) – Old generation, not recommended for new apps. Performs routing at Layer 4 and Layer 7.



Application Load Balancer (ALB)

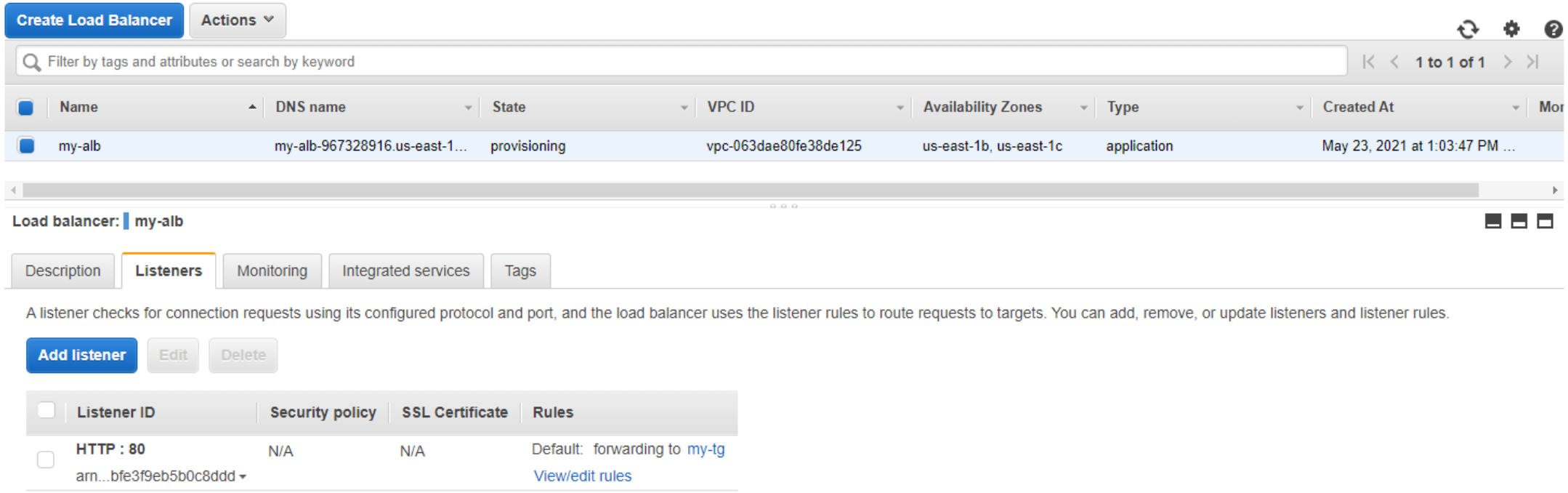
A load balancer serves as the single point of contact for clients. The load balancer distributes incoming application traffic across multiple targets, such as EC2 instances, in multiple Availability Zones. This increases the availability of your application. You add one or more listeners to your load balancer.



ALB Listener

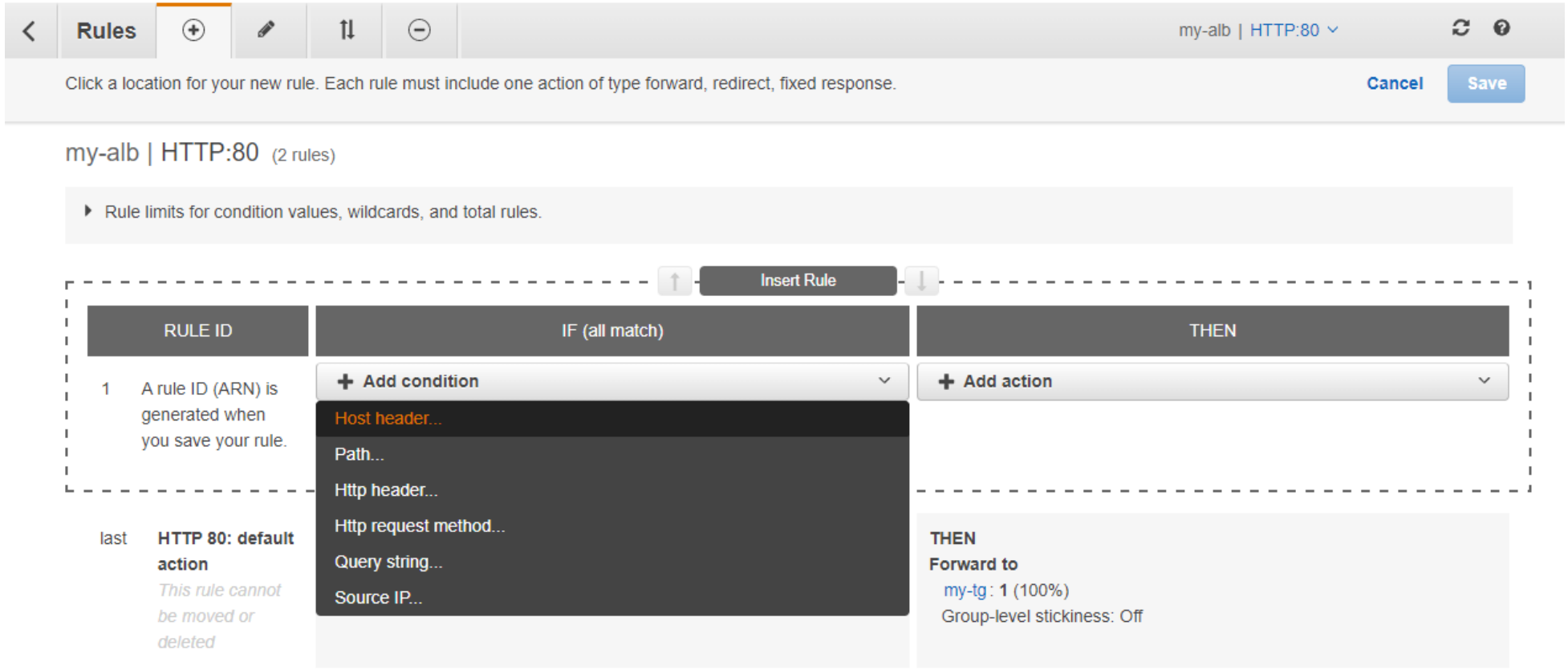
A listener checks for connection requests from clients, using the **protocol** or **port** that you configure. ALB supports HTTP (80) and HTTPS (443). That means you can have 2 rules on ALB at most.

A **certificate** is attached to the listener. You must define a default certificate if using HTTPS. AWS and custom certificates are stored on Amazon Certificate Manager (ACM). AWS certificates are free!



ALB Listener Rules

The **rules** that you define for a listener determine how the load balancer routes requests to its **registered targets**. Each rule consists of a **priority, actions, conditions**.



ALB Listener Rules

When the conditions for a rule are met, then its actions are performed. You must define a default rule for each listener, and you can optionally define additional rules. There could be many apps behind an ALB. You can have the following rules:

- If the request path contains “hr”, then forward them to the service for the HR module.
- If the request path contains “finance”, then forward requests to the target group that has the Finance app.

Not only the request path, you can also define rules based on query strings, headers, source IP, HTTP methods. And route requests to different target groups (or apps) based on the rules. Try out the listener rules on the AWS console.

Running multiple apps behind a single ALB

- You can run multiple apps behind an ALB. For that, you need to write listener rules and create a TG for each app.
- Not only the request path, you can define rules based on other parameters such as headers, query strings.

Request	Rule	Target Group
example.com/app1	Path based, if the request includes “/app1” in the path, route to the “TG1”.	TG1 – Target Group that groups resources (such as EC2 instances, ECS tasks, Lambda instances) that run the “app1”.
example.com/app2	Path based, if the request includes “/app2” in the path, route to the “TG2”.	TG2 – Target Group that groups resources (such as EC2 instances, ECS tasks, Lambda instances) that run the “app2”.

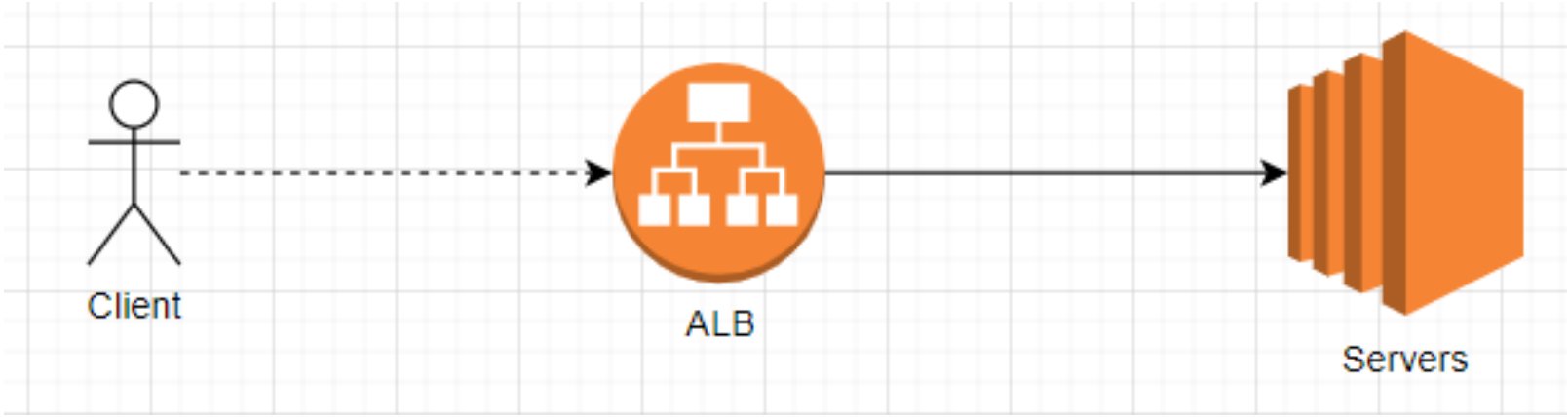
ALB Target Groups

ALB target group is a logical grouping of targets, usually Amazon EC2 instances or ECS services, that receive traffic from the ALB based on rules defined in the listener configuration. When you create an ALB, you configure listeners to process incoming connections and forward requests to one or more target groups. Key concepts include:

1. **Targets**: EC2 instances, ECS tasks, or IP addresses, that receive traffic from the ALB.
2. **Health Checks**: ALB continuously monitors the health of targets. If a target fails the health check, the ALB stops sending traffic to that target until it passes the health check again. Increasing reliability.
3. **Routing**: Target groups enable routing of requests to different sets of targets based on rules defined in the listener configuration. This allows you to route traffic to different target groups based on factors like path patterns, hostnames, or query parameters.
4. **Port Configuration**: Each target group is associated with a specific port on the targets.
5. **Attributes**: Target groups have attributes such as stickiness settings, which enable session persistence by directing subsequent requests from a client to the same target.

ALB Target Groups

Each target group routes requests to one or more registered targets, such as EC2 instances, **using the protocol and port number that you specify**. That means there are 2 connections on the way. One from the client to ALB and then another one from the ALB to the registered target (servers) in the TG.



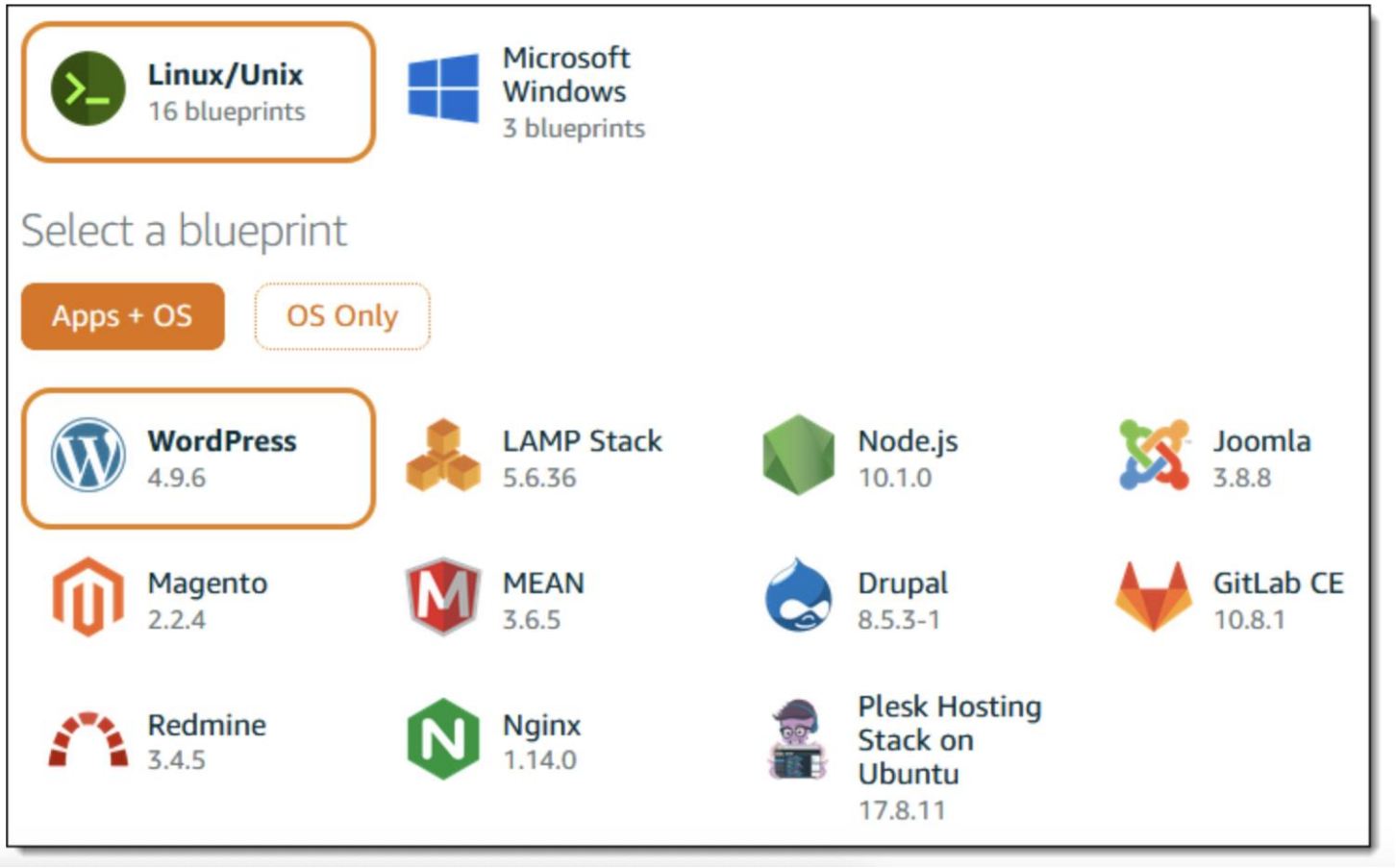
ALB Target Groups

In most cases, the Target Group (TG) is multiple resources that is running the same app. It could be a fleet of EC2 servers running the same app based on the custom AMI. Resources in a TG are mostly created by Auto Scaling Groups (ASG) and the number of resources in the TG increases and decreases automatically based on the incoming traffic.

You can configure health checks on a per-target group basis. Health checks are performed on all targets registered to a target group.

Note: When you associate a resource with an ALB, you associate it with its target group. When creating an ALB, the target group can have no resources. But the important thing is you must specify the right target group type (IP, instance, lambda).

Amazon Lightsail



Amazon Lightsail

- It is a simple VPS (Virtual Private Server)
- Small businesses, developers and students need a simpler, more predictable pricing model

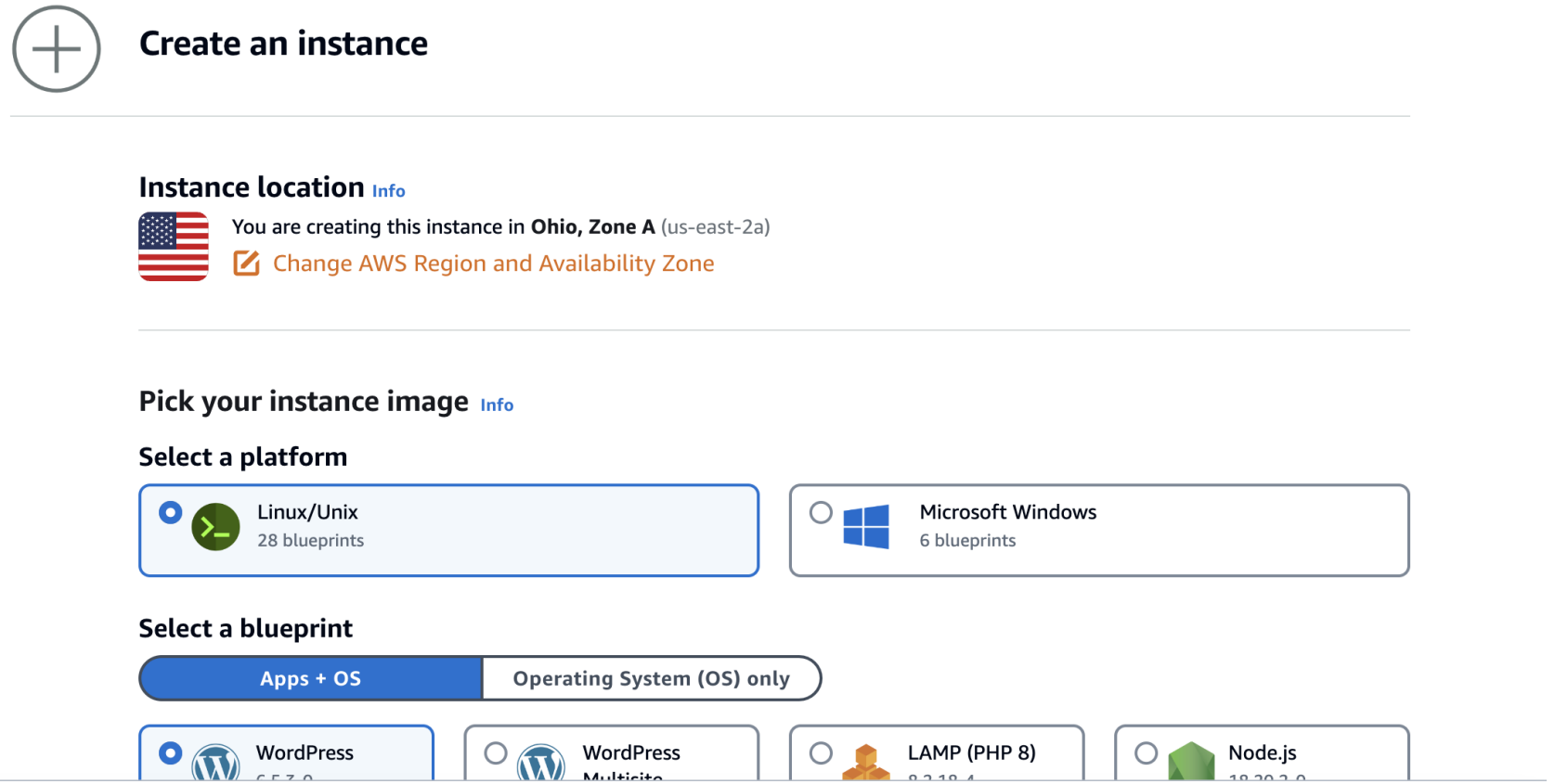
Amazon Lightsail

- Key features
  - Simplified VPS: streamlined for easy deployment.
  - Pricing: Fixed monthly cost that includes a virtual server, SSD-based storage, data transfer, DNS management, and static IP.
  - Bundled Ecosystem: Can easily add on databases, networking features and storage without needing deep AWS expertise.

Amazon Lightsail

- Lightsail vs. EC2
  - Simplicity
  - Control

Amazon Lightsail



References

- https://docs.aws.amazon.com/
- ChatGPT: https://chatgpt.com/
- Google AI: https://gemini.google.com/app
- https://dev.to/aws-builders/all-you-need-to-know-about-aws-compute-services-17gf